

Common-root Cartan quotient, rational Taylor-owner subdivision, and smart-path boundary

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-103 closed the complete regular H_N/REG packet, and R-104 proved that the complete endpoint-owner algebra can be reconstructed without defect on each temporally faithful fixed source chart. The remaining analytic theorem is nevertheless the uniform lower bound for the complete source action, which R-093 identifies exactly with $\text{OVERLAP}_{\text{src}}$ and the $q = 10/9$ Nelson bound. This note returns to the two historical R-085 atoms and asks which parts survive representation-preserving subdivision. It also tests the most economical static continuation, the A9 smart path.

The outcome has four parts.

1. Complete common-root Cartan edges have an exact signed-first endpoint quotient. Any two complete-heat subdivisions with the same extended endpoints give the same grouped square. A closed extended loop gives zero grouped charge, even when the sum of edgewise squares is positive.
2. The historical rational form $F_{6.5}$ and the later fixed-chart coordinate $\mathcal{K}_R$ do not descend to the subdivision quotient. An exact production-fibre subdivision changes the sign of both. The complete raw-Wick endpoint increment ΔW_R is unchanged because the labelled-owner defects cancel exactly.
3. Positive semidefiniteness, divergence freedom, quadratic coefficient growth, and sextic coercivity do not imply A9 smart-path monotonicity. A two-dimensional exact fixture has strictly negative first variation, and independent block copies amplify it linearly.
4. The natural all-finite-entropy pointwise relative-bracket theorem is a sufficient route to Nelson but is itself false. One exact production top-shell ray and countably many translated Gaussian laws force its putative coefficient to satisfy $b(t) \geq 3/t$. Only a Gibbs-law-specific or time-integrated signed A9 estimate survives; invoking the complete source-action lower bound directly would simply assume Nelson.

The first two results are finite-cutoff, fixed-positive-floor algebraic statements in the A1 production frame. The Cartan quotient uses one fixed Malliavin root and the complete value-plus-heat edge. The rational fixture uses the exact active scalar production Gram and one common target heat. The A9 fixture is a structural method no-go, not a production counterexample. No uniform $\text{OVERLAP}_{\text{src}}$ bound, Nelson estimate, cutoff or floor removal, interacting measure, or Sector-A closure is claimed. Tier stays T4.

2 Complete common-root Cartan edges

Fix one Malliavin root j , one spatial output m , and one OU parameter t . Write

$$\mathcal{L}_{j,m,t} := \Pi_m P_t^{(j)} \mathbb{D}_j. \tag{2.1}$$

The point is that $\mathcal{L}_{j,m,t}$ is linear and remains fixed during a pure subdivision at that root. Let

$$(\Sigma_0, z_0), (\Sigma_1, z_1), \dots, (\Sigma_L, z_L) \quad (2.2)$$

be a finite path in the extended value-heat state. Its complete edge is

$$\begin{aligned} E_\ell &= J_{\Sigma_\ell}(z_\ell) - J_{\Sigma_\ell}(z_{\ell-1}) \\ &\quad + J_{\Sigma_\ell}(z_{\ell-1}) - J_{\Sigma_{\ell-1}}(z_{\ell-1}). \end{aligned} \quad (2.3)$$

The first line is the value edge. The second is the heat compensator and is part of the complete R-099 coordinate.

Theorem 2.1 (complete common-root endpoint quotient)

Pathwise,

$$\boxed{\sum_{\ell=1}^L \mathcal{L}_{j,m,t} E_\ell = \mathcal{L}_{j,m,t} \{J_{\Sigma_L}(z_L) - J_{\Sigma_0}(z_0)\}.} \quad (2.4)$$

Consequently,

$$\begin{aligned} &\int_0^\infty e^{-2t} \mathbb{E} \left\| \sum_\ell \mathcal{L}_{j,m,t} E_\ell \right\|_{\text{HS}}^2 dt \\ &= \int_0^\infty e^{-2t} \mathbb{E} \left\| \mathcal{L}_{j,m,t} [J_{\Sigma_L}(z_L) - J_{\Sigma_0}(z_0)] \right\|_{\text{HS}}^2 dt. \end{aligned} \quad (2.5)$$

Proof. The two lines of (2.3) cancel at their common middle value and give

$$E_\ell = J_{\Sigma_\ell}(z_\ell) - J_{\Sigma_{\ell-1}}(z_{\ell-1}). \quad (2.6)$$

Summation telescopes in the extended state. Apply the fixed linear map $\mathcal{L}_{j,m,t}$, then square only after summation. Integration and expectation preserve the exact identity. $\square$

Two consequences are immediate. Every complete common-root partition with the same extended endpoints has the same grouped energy. If the endpoints agree, its grouped charge and grouped energy are exactly zero. In contrast,

$$\sum_\ell \|\mathcal{L} E_\ell\|^2 \quad (2.7)$$

can be strictly positive on a closed loop. Thus (2.7) is a representation artefact, not an invariant energy coordinate. The executable uses a nonlinear two-component current and an exact closed loop for which the grouped square is zero and the edgewise square is 62132.

Why the heat compensator is mandatory

Keeping only the first line of (2.3) gives

$$\begin{aligned} S_{\text{src}} &= J_{\Sigma_L}(z_L) - J_{\Sigma_0}(z_0) \\ &\quad - \sum_\ell [J_{\Sigma_\ell} - J_{\Sigma_{\ell-1}}](z_{\ell-1}). \end{aligned} \quad (2.8)$$

The last sum is a genuine defect. The quotient theorem is therefore a complete-heat theorem, not a value-edge theorem.

3 Relation to the R-085/R-088 Cartan ledger

The normalized R-084 target audited in R-088 has no outer 2^j weight. Its direct Schur coordinate is

$$\mathcal{S}_C = \sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \left\| Q_{j,C} \sum_{k \leq j} v_{A,j,k,t} \right\|_{\text{HS}}^2 dt. \quad (3.1)$$

For a complete common-root subdivision, the correct coarse atom is

$$\bar{v}_{A,j,k,t} := \sum_{\ell \in I_k} \mathcal{L}_{j, \cdot, t} E_\ell, \quad (3.2)$$

and one q_k is attached to the coarse endpoint class. Introducing a separate nonnegative $q_{k,\ell}$ for every artificial edge changes the coordinate and can recreate the R-098 A^2 loss.

Corollary 3.1 (lossless regular transport through pure subdivision)

If an R-092 regular chart is changed only by a representation-preserving complete common-root subdivision, with no new root, reveal, control, or extended endpoint, its accepted regular Cartan estimate is unchanged:

$$H_C^{\text{reg}}(C) \leq C_{e,L} 2^{-(C-5)/2} \{C_{\varepsilon,\delta,e,L} + \varepsilon \mathbb{E}X + \delta \mathbb{E}Y\}. \quad (3.3)$$

This is a transport of the integrated R-092 theorem, not a proof of literal R-085 (4.11). Distinct temporal roots carry different $\mathbb{D}_j P_t^{(j)}$ maps and cannot be combined by (2.4). R-099's fresh-root inverse-visit fixture therefore remains outside the theorem and still rules out the arbitrary-progressive mixed-only extension of (4.11).

4 The exact rational ownership correction

Let $B_0 = B_R(u)$ and $B_1 = B_R(u+a)$. Define the second Taylor polynomial and remainder by

$$B_T = B_0 + DB_0[a] + \frac{1}{2}D^2B_0[a,a], \quad L = B_1 - B_T. \quad (4.1)$$

For a derivative value G , derivative shift c , and covariance Γ , put

$$Q = G^{\otimes 2} - \Gamma, \quad Q^a = (G+c)^{\otimes 2} - \Gamma. \quad (4.2)$$

The three R-101 complete-owner labels are

$$\mathcal{R}_Q = \frac{1}{2}(B_1 - B_0) : Q, \quad \mathcal{M}_U = G^T B_T c, \quad (4.3)$$

$$\mathcal{K}_R = G^T L c + \frac{1}{2}c^T B_1 c. \quad (4.4)$$

Theorem 4.1 (Taylor-owner correction and complete endpoint)

The historical R-085 form satisfies

$$\boxed{F_{6.5} = \frac{1}{2}L : Q^a + \frac{1}{2}c^T B_T c = \frac{1}{2}L : Q + \mathcal{K}_R.} \quad (4.5)$$

The nonduplicating complete R-101 replacement is

$$\boxed{\Delta W_R = \frac{1}{2}B_1 : Q^a - \frac{1}{2}B_0 : Q = \mathcal{R}_Q + \mathcal{M}_U + \mathcal{K}_R.} \quad (4.6)$$

Proof. Insert $Q^a = Q + G \otimes c + c \otimes G + c \otimes c$ into the first expression in (4.5), use symmetry of L , and substitute $B_1 = B_T + L$. For (4.6), expand the endpoint difference and group first the Q coefficient change, then the Taylor mixed term, and finally the remainder mixed term together with the complete shifted square. $\square$

Thus $L : Q/2$ is already inside the indivisible raw-Wick owner $\mathcal{R}_Q$. Retaining it in $F_{6.5}$ while also using the R-101 split duplicates ownership. The one-chart identity (4.5) remains correct; its use as a separate progressive owner is what fails below.

5 Rational subdivision no-go

On the active scalar production ray, normalize the positive coefficient c_1 and the floor e to one. The exact Gram coefficient is

$$b(x) = \frac{4x^2(4x^2 + 9)^2}{81(1 + x^2)^2}. \quad (5.1)$$

Its exact jets are

$$b(0) = b'(0) = 0, \quad b''(0) = 8, \quad (5.2)$$

$$b(1) = \frac{169}{81}, \quad b'(1) = \frac{208}{81}, \quad b''(1) = -\frac{2}{81}, \quad b(2) = \frac{400}{81}. \quad (5.3)$$

Hence

$$L(0, 2) = -\frac{896}{81}, \quad L(0, 1) = -\frac{155}{81}, \quad L(1, 1) = \frac{8}{27}. \quad (5.4)$$

Take $\Gamma = 1$. Compare the single chart

$$(u, G) = (0, 1), \quad (a, c) = (2, 2) \quad (5.5)$$

with the pure representation-preserving subdivision

$$(0, 1) \xrightarrow{(1,1)} (1, 2) \xrightarrow{(1,1)} (2, 3). \quad (5.6)$$

Proposition 5.1 (labelled rational owners do not descend)

The exact values are

quantity	one chart	two-step sum	one minus sum
$\mathcal{R}_Q$	0	77/18	-77/18
$\mathcal{M}_U$	32	1076/81	1516/81
$\mathcal{K}_R$	-992/81	355/162	-2339/162
$F_{6.5}$	-992/81	427/162	-2411/162
ΔW_R	1600/81	1600/81	0

(5.7)

In particular, both $F_{6.5}$ and $\mathcal{K}_R$ change sign under the subdivision, while

$$\text{defect}(\mathcal{R}_Q) + \text{defect}(\mathcal{M}_U) + \text{defect}(\mathcal{K}_R) = 0. \quad (5.8)$$

Proof. Substitute (5.2)–(5.4) into (4.3)–(4.6). For example, the one-chart values are

$$B_T = 16, \quad Q = 0, \quad \mathcal{M}_U = 32, \quad \mathcal{K}_R = F_{6.5} = -\frac{992}{81}. \quad (5.9)$$

The two exact step expansions give the remaining entries of (5.7). Equation (4.6) proves (5.8) and the endpoint equality. Both the symbolic certificate and a non-importing standard-library rational-jet certificate derive every fraction independently. $\square$

The fixture is not an artefact of unit floor or covariance. Since

$$B_e(\sqrt{e}s) = c_1 e b(s), \quad (5.10)$$

scaling value jets by $\sqrt{e}$, derivative jets by $\sqrt{\Gamma_0}$, and taking covariance $\Gamma_0 > 0$ multiplies every entry of (5.7) by the same positive number $c_1 e \Gamma_0$. The signs and zero endpoint defect persist.

This is a production-fibre owner-coordinate no-go, not a globally realized torus control. It does not refute R-102's fixed regular no-revisit $\mathcal{K}_R$ estimate. It proves only that neither $F_{6.5}$ nor $\mathcal{K}_R$ may be temporalized visit by visit. Across changing target heats, the invariant coordinate is the complete extended R-100/R-101 row in its stated conditional measurability scope; no pathwise additivity across different random heat operators is asserted.

6 A9 smart-path audit

Let $q = 10/9$. At the A9 interpolation time t , write

$$G_{J,t} := -W_{J,t}/q, \quad \mathcal{F}_{J,t}(\nu) := H(\nu \mid \gamma_J) + q\mathbb{E}_\nu G_{J,t}. \quad (6.1)$$

For the exact Gibbs tilt $\nu_{J,t}$,

$$\mathcal{F}_{J,t}(\nu_{J,t}) = -\Phi_{J,t}. \quad (6.2)$$

R-104 reconstructs the endpoint complete owner, but this supplies only the same variational action:

$$\mathcal{F}_{J,0} = q\mathcal{A}_J^{\text{own}} = q\mathcal{A}_J, \quad \mathcal{A}_J^{\text{phys}} = \mathcal{A}_J + \mathfrak{D}_{\text{CM}}, \quad \mathfrak{D}_{\text{CM}} \geq 0. \quad (6.3)$$

R-093 then gives

$$\inf_{\mathfrak{Z}_J} \mathcal{A}_J = -\frac{9}{10} \log \mathbb{E} \exp\{-\frac{10}{9} G_J(X_J)\}. \quad (6.4)$$

Using a uniform lower bound for the left side inside the smart path is therefore circular: it is exactly Nelson.

There is also a centering obstruction at intermediate time. If a packet M obeys $\mathbb{E}_\gamma[M \mid \mathcal{F}] = 0$, then

$$\mathbb{E}_{\nu_t} M = \frac{1}{Z_t} \mathbb{E}_\gamma \text{Cov}_\gamma(M, e^{W_t} \mid \mathcal{F}). \quad (6.5)$$

The same-root resolvent heat

$$S_t(x) = T(x)(I + qtT(x))^{-1} \quad (6.6)$$

cannot be frozen using R-104's predictable fresh-root-independent heat scope. The A9 coefficient and divergence corrections are exactly the missing covariance terms.

An exact generic monotonicity no-go

On $\mathbb{R}^2$, set $U(x) = \lambda|x|^6$ and

$$T_c(x) = cT_0(x), \quad T_0(x) = \begin{pmatrix} x_1^2 + 3x_2^2 & -2x_1x_2 \\ -2x_1x_2 & 3x_1^2 + x_2^2 \end{pmatrix}. \quad (6.7)$$

Direct algebra gives

$$\text{spec } T_0 = \{|x|^2, 3|x|^2\}, \quad \text{div } T_0 = 0, \quad T_0 x = |x|^2 x, \quad (6.8)$$

$$x^T T_0 x = |x|^4, \quad \text{Tr } T_0 = 4|x|^2. \quad (6.9)$$

Because $\log \det_2$ has no term linear in c ,

$$\partial_c(\Phi_{c,1} - \Phi_{c,0})|_{c=0} = \frac{q}{2} \mathbb{E}_\mu(|X|^4 - 4|X|^2), \quad (6.10)$$

where $d\mu \propto e^{-q\lambda|x|^6}d\gamma_2$. Put $Y = |X|^2/2$ and $a = 8q\lambda$. Its density is proportional to e^{-y-ay^3} . Integration of $(y^2e^{-y-ay^3})'$ gives

$$\mathbb{E}(Y^2) - 2\mathbb{E}Y = -3a\mathbb{E}Y^4 < 0. \quad (6.11)$$

At $q = 10/9$ and $\lambda = 3/20$, $a = 4/3$ and (6.10) equals

$$-\frac{80}{9}\mathbb{E}_\mu Y^4 < 0. \quad (6.12)$$

Independent block copies multiply this first variation by the number of blocks. Thus PSD, divergence freedom, quadratic growth, and a positive sextic do not yield cutoff- or dimension-uniform A9 monotonicity. This is not a production counterexample; additional production owner/covariance structure could still prove a relative estimate.

Conditional implication 6.1 (relative-bracket Gronwall route)

Let $\mathfrak{B}_{J,t}$ denote the full exact A9 bracket. A sufficient new theorem would produce cutoff-independent integrable $a, b \geq 0$ such that, for every finite-entropy ν and almost every t ,

$$\frac{q}{2}\mathbb{E}_\nu \mathfrak{B}_{J,t} + b(t)\mathcal{F}_{J,t}(\nu) \geq -a(t). \quad (6.13)$$

At $\nu = \nu_{J,t}$ this becomes

$$\Phi'_{J,t} - b(t)\Phi_{J,t} \geq -a(t). \quad (6.14)$$

The integrating factor gives

$$\Phi_{J,0} \leq e^{-\int_0^1 b} \Phi_{J,1} + \int_0^1 e^{-\int_0^s b} a(s) ds. \quad (6.15)$$

The independent endpoint is already controlled, so (6.13) would close Nelson. This is a sufficient implication, not an equivalence. The next proposition shows that its quantifier over every finite-entropy law is too strong.

Proposition 6.2 (production top-shell all-law no-go)

There are no cutoff-independent $a, b \in L^1(0,1)$, $a, b \geq 0$, for which (6.13) holds for every finite-entropy law at almost every time.

Proof. It is enough to work at one fixed finite cutoff. For the admitted sharp-cube family $Q_J = \{n : |n|_\infty \leq 2^J\}$, fix $J \geq 1$, take $n = (2^{J-1}, 0, 0)$ and $k = 2\pi n/L$, and put $g(x) = \cos(k \cdot x)$. Then k is retained while $3k$ and $5k$ are removed. Let $u = e_{1,R}$ be a real unit active-doublet horizontal vector and choose the white-coordinate ray $e = K_J^{-1}(gu)$; this is legitimate because the pinned finite-cutoff A1 covariance is positive definite. With $P = 4 + 10^{-12}$ and the exact production coefficient

$$a_{\text{prod}} = \frac{9}{500P} > 0,$$

the production Fierz formula gives, uniformly on the amplitude ray,

$$B(Agu)u = 4a_{\text{prod}}A^2g^2u + O(1), \quad a_{\text{prod}} > 0. \quad (6.16)$$

The two top-shell identities are

$$P_J g^3 = \frac{3}{4}g, \quad P_J g^5 = \frac{10}{16}g, \quad (6.17)$$

and

$$P_J D^*(g^2 Dg) = \frac{|k|^2}{4}g. \quad (6.18)$$

If T_∞ denotes the A^2 -leading coefficient of $T(Ae)$, the A1 positivity and (6.16)–(6.18) give

$$T(Ae) = A^2 T_\infty + O(1), \quad T_\infty \succeq 0, \quad T_\infty e = a_{\text{prod}} |k|^2 K_J^*(gu) \neq 0. \quad (6.19)$$

The projected sextic gradient $d := DU_6(e)$ is a positive multiple of this same vector $T_\infty e$, hence $d \in \text{Ran } T_\infty$. Consequently, if P_∞ is the orthogonal projection onto $\text{Ran } T_\infty$ and

$$U_6(Ae) = u_6 A^6, \quad u_6 = \frac{5\gamma L^3}{96} > 0, \quad (6.20)$$

then Euler homogeneity gives

$$\langle d, P_\infty e \rangle = \langle DU_6(e), e \rangle = 6u_6. \quad (6.21)$$

The finite-dimensional range–kernel block decomposition and $S_t = (qt)^{-1} \{I - (I + qtT)^{-1}\}$ now yield, for every fixed $t > 0$,

$$\frac{qt}{A} S_t(Ae) \longrightarrow P_\infty e. \quad (6.22)$$

There is no hidden contribution at sextic order. At fixed t , $\|S_t(Ae)\| = O_t(A)$ and $\|T_i(Ae)\| = O(A)$. Formula A9-(3.5) gives $\|a_t(Ae)\| = O_t(A^3)$, so $\langle a_t, S_t(Ae) \rangle = O_t(A^4)$; likewise $\langle \text{div } S_t, Ae \rangle = O_t(A^2)$. All quartic, determinant, trace, and covariance-normal terms are $O_t(A^4)$. Thus only the sextic score contributes at order A^6 .

Take $A_n = 2^n$ and the countable family of finite-entropy laws $\nu_n = N(A_n e, A_n^{-40} I)$. Intersect the full-measure time sets on which (6.13) holds for these laws. At every remaining fixed t , concentration, (6.21), and (6.22) give

$$\mathcal{F}_{J,t}(\nu_n) = qu_6 A_n^6 + o(A_n^6), \quad (6.23)$$

$$\frac{q}{2} \mathbb{E}_{\nu_n} \mathfrak{B}_{J,t} = -\frac{3qu_6}{t} A_n^6 + o(A_n^6). \quad (6.24)$$

The relative entropy is $O(A_n^2 + \log A_n)$ and the shrinking covariance makes the concentration errors $o(A_n^6)$. Divide (6.13) by $qu_6 A_n^6$ and let $n \rightarrow \infty$. Since $a(t)$ is finite almost everywhere,

$$b(t) \geq \frac{3}{t} \quad \text{for almost every } t, \quad (6.25)$$

contradicting $b \in L^1(0, 1)$. □

Proposition 6.2 does not test the actual Gibbs law $\nu_{J,t}$, whose coefficient correlations are load-bearing, and it does not test an estimate integrated in t before absolute values. A Gibbs-law-only or time-integrated signed bracket is therefore the surviving A9 form. R-104 still cannot freeze the same-root resolvent heat.

7 Deterministic extraction boundaries

The translation authority already rules out a cutoff-uniform pathwise remainder. At the Gaussian-null field $X = 0$ and constant active-doublet shift $h = te_1$,

$$V_J^{\text{ren}}(te_1) = -\mathcal{C}_J(te_1), \quad \frac{\mathcal{C}_J(te_1)}{Jt^2 L^3} \longrightarrow \kappa = 0.001248334393361145 \dots \quad (7.1)$$

Even using the entire R-104 source and sextic budgets $\eta = 9/20$, $\zeta = 3/20$, the large- J paid polynomial has the form

$$P_J(t) = L^3 \{-d_J t^2 + \zeta t^6\}, \quad d_J \sim \kappa J. \quad (7.2)$$

Its minimizer satisfies $t^4 = d_J/(3\zeta)$ and

$$\inf_t P_J(t) = -\frac{2}{3\sqrt{3}\zeta} L^3 d_J^{3/2} = -\frac{4\sqrt{5}}{9} L^3 d_J^{3/2} \longrightarrow -\infty. \quad (7.3)$$

Since $X = 0$ is Gaussian-null, this is a method no-go rather than a Nelson counterexample. It proves that expectation-level Wick cancellation is indispensable.

There is a second exact critical-homogeneity boundary. Suppose absolute value extraction produces

$$CRE^{3/4}Y^{1/4} \leq \eta E + \zeta Y + F(R) \quad (7.4)$$

for all $E, Y \geq 0$. Put $Y/E = \eta/(3\zeta)$ and let E grow. A finite $F(R)$ exists only if

$$CR \leq \frac{4\eta^{3/4}\zeta^{1/4}}{3^{3/4}}. \quad (7.5)$$

With the full budgets the threshold is exactly $3/5$. A Gaussian current norm is unbounded, so extracting it before the signed expectation cannot leave a finite universal remainder. The analogous half-half threshold is $2\sqrt{\eta\zeta} = 3\sqrt{3}/10$.

The viable frozen branch is different. If B is strict-past measurable and Y_k is a fresh Gaussian derivative shell with covariance Γ_k , then

$$\begin{aligned} \log \mathbb{E} \left[e^{-\frac{q}{2}(Y_k^T B Y_k - \text{Tr}(B \Gamma_k))} \mid \mathcal{F}_{k-1} \right] \\ = -\frac{1}{2} \log \det_2(I + q\Gamma_k^{1/2} B \Gamma_k^{1/2}) \leq \frac{q^2}{4} \|\Gamma_k^{1/2} B \Gamma_k^{1/2}\|_{\mathfrak{S}_2}^2. \end{aligned} \quad (7.6)$$

In three dimensions the frozen cost is quartic and sextic-subcritical. The unresolved part is exactly the same-root coefficient increment. Equations (7.3)–(7.5) say it must remain inside a centered signed Wick/current/square packet rather than another pathwise Young estimate.

8 Exact one-Fourier-pair theorem and cross-mode boundary

There is no divergence already inside one self-coupled Fourier pair. On a normalized one-dimensional orbit, let

$$X = A + \sigma\{(\xi + h_1) \cos(kx) + (\eta + h_2) \sin(kx)\}, \quad (8.1)$$

where ξ, η are independent standard Gaussians and h is measurable before this pair is revealed. In the scalar quadratic coefficient model $B(z) = \beta z^2$, put

$$S_h = (\xi + h_1)^2 + (\eta + h_2)^2, \quad \Gamma = k^2 \sigma^2. \quad (8.2)$$

Exact trigonometric averaging of the covariance-normal energy gives

$$V = \alpha(S_h - 2) + \mu(S_h^2 - 4S_h), \quad (8.3)$$

$$\alpha = \frac{\beta k^2 A^2 \sigma^2}{4}, \quad \mu = \frac{\beta k^2 \sigma^4}{16}. \quad (8.4)$$

Theorem 8.1 (uniform predictable-source bound for one pair)

For every $q > 0$ and every strict-past value of h ,

$$\log \mathbb{E}_{\xi, \eta} e^{-qV} \leq 4q\mu + t - \log(1+t) \leq 4q\mu + \frac{t^2}{2}, \quad t = 2q\alpha. \quad (8.5)$$

Proof. Since $S_h^2 - 4S_h = (S_h - 2)^2 - 4 \geq -4$,

$$-qV \leq 4q\mu + 2q\alpha - q\alpha S_h. \quad (8.6)$$

For $a \geq 0$, completing the square in two independent Gaussians gives

$$\mathbb{E} e^{-a|\zeta+h|^2} = \frac{1}{1+2a} \exp \left\{ -\frac{a|h|^2}{1+2a} \right\} \leq \frac{1}{1+2a}. \quad (8.7)$$

Take $a = q\alpha$. Finally, $t - \log(1+t) \leq t^2/2$ because the difference is zero at $t = 0$ and has derivative $t^2/(1+t) \geq 0$. $\square$

For production covariance $\sigma_k^2 \lesssim \langle k \rangle^{-4}$ in three dimensions,

$$\sum_k \mu_k \lesssim \sum_k \langle k \rangle^{-6} < \infty, \quad \sum_k t_k^2 \lesssim A^4 \sum_k \langle k \rangle^{-4} < \infty. \quad (8.8)$$

The A^4 background cost is sextic-subcritical. Thus the artificial mode-diagonal quadratic model has a cutoff-uniform conditional source bound. Its pathwise lower cost would instead be proportional to $\sum_k \langle k \rangle^{-2}$; expectation changes the linear determinant loss into the summable quadratic remainder $t - \log(1+t)$.

The physical field is not a product of these pair models. For

$$X = A + r \cos x + u \cos 2x, \quad (8.9)$$

direct averaging gives

$$\langle X^2 |\partial X|^2 \rangle = \frac{4A^2 r^2 + 16A^2 u^2 + 12Ar^2 u + r^4 + 10r^2 u^2 + 4u^4}{8}. \quad (8.10)$$

Relative to the two isolated diagonal raws, the cross-mode resonance is

$$\frac{r^2 u (6A + 5u)}{4}. \quad (8.11)$$

It is sign-indefinite; $A = 1, u = -1$ gives $-r^2/4$. Frequencies $k, 2k$ multiply (8.10)–(8.11) by k^2 . The complete raw polynomial remains nonnegative, so this is not a negative full-action direction. It proves that the pair bounds cannot simply be multiplied for the physical field. The unresolved target is cross-mode/forest tilted-law control while keeping the complete covariance-normal packet intact.

9 Attempt, failure, and evidence map

Route	Verdict	Reusable reason
Complete common-root Cartan subdivision	advanced	Signed edges telescope to the extended endpoint before the square; no multiplicity loss.
Edgewise nonnegative Cartan ledger	failed	Closed complete loops have zero grouped charge but positive edgewise square.
Historical $F_{6.5}$ as progressive owner	superseded	Exact production subdivision changes its sign while preserving the complete endpoint.
$\mathcal{K}_R$ visit by visit	failed	The same fixture changes its sign; R-102 remains valid only on its declared fixed regular chart.
Complete rational endpoint	advanced algebraically	The three labelled defects cancel exactly in ΔW_R .
Generic A9 monotonicity	failed	A PSD divergence-free sextic fixture has negative first variation and block-linear amplification.
R-104 at the A9 endpoint	circular	Its complete action is exactly the R-093 Nelson variational objective.
Pathwise coercivity or critical Young	failed	The Gaussian-null constant mode diverges as $-J^{3/2}$; critical homogeneous extraction has a finite threshold but the Gaussian norm is unbounded.
All-law pointwise relative A9 bracket	failed	The production top-shell ray forces $b(t) \geq 3/t$ after a countable translated-law limit.
Gibbs-only or time- integrated A9 bracket	open	Must retain the full same-root production correlations and signed time cancellation.
One Fourier pair with predictable source	advanced	Exact noncentral Gaussian integration converts the nonsummable linear cost to summable $t - \log(1 + t)$ and leaves only a summable quartic self-pair cost.
Product of one-pair bounds for the physical field	failed	Cross-mode resonances are sign-indefinite and prevent factorization.

The evidence executables derive the production rational fractions, Cartan quotient fixtures, A9 polynomial identities, full-budget thresholds, and scope firewalls. The primary route uses symbolic rational algebra. The independent route uses only standard-library rational two-jets and a separate bivariate polynomial implementation. Neither imports the other.

10 Devil’s-advocate review

1. **DISMISSED – the Cartan theorem proves R-085 (4.11).** It proves an endpoint quotient and transports the already accepted R-092 regular theorem through pure common-root subdivision. It creates no new q_k ledger and does not cover distinct roots.
2. **DISMISSED – deleting the heat compensator preserves the telescope.** Equation (2.8) gives the exact nonzero defect. The theorem is explicitly complete-heat only.
3. **DISMISSED – the rational sign flip falsifies R-102.** R-102 is a fixed regular no-revisit estimate. The fixture falsifies only its termwise temporalization under an artificial subdivision.

4. **VALID-with-mitigation – the rational fixture is only local.** It is stated as a production-fibre owner-coordinate no-go, not as a globally realized torus control or action counterexample.
5. **DISMISSED – generic A9 monotonicity follows from PSD and the sextic.** Equations (6.7)–(6.12) give a strict exact counterexample. A Gibbs-specific or time-integrated production bracket remains possible.
6. **DISMISSED – R-104 supplies that relative-bracket theorem.** It supplies the complete endpoint coordinate at $t = 0$. The intermediate same-root resolvent heat and coefficient covariance are outside its predictable-heat scope.
7. **DISMISSED – the pathwise constant-mode fixture refutes Nelson.** The tested base field has Gaussian probability zero. The fixture refutes only deterministic pointwise coercivity.
8. **DISMISSED – the all-law bracket no-go refutes Nelson.** It does not. Proposition 6.2 tests arbitrary translated laws, not the actual Gibbs tilt, and leaves time-integrated signed cancellation open.
9. **DISMISSED – the final source-action lower bound is now proved.** It is not. R-093/R-104 identify it exactly with Nelson; Proposition 6.2 only retires one overstrong sufficient route.

11 Consequences and exact frontier

R-105 closes the representation audit that remained around the historical R-085 atoms. The Cartan common-root subdivision artefact is removed exactly. The old rational $F_{6.5}$ is retained only as a one-chart identity and is superseded as a load-bearing progressive owner. The R-101 complete endpoint is the only subdivision-stable rational coordinate in the stated common-heat scope. R-103 regular H_N/REG and R-104 fixed-chart assembly are unchanged.

The smallest live theorem is now unambiguous:

$$\inf_{(S_\pi, h) \in \mathfrak{T}_J} \left[\mathbb{E} G_J(S_\pi(\xi + h)) + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2 \right] \geq -C \quad \text{uniformly in } J. \quad (11.1)$$

By R-093 this is exactly $\text{OVERLAP}_{\text{src}}$ and exactly the $q = 10/9$ Nelson bound. No post-assembly intermediate gate remains. The two honest continuations are either a complete signed same-root packet estimate for (11.1), or a Gibbs-law-specific/time-integrated A9 estimate. Repeating common-root subdivision, $F_{6.5}$, visitwise $\mathcal{K}_R$, generic A9 monotonicity, the all-law pointwise bracket (6.13), or pathwise Young extraction cannot close the theorem.

12 Result footer

Result ID: A13-CLASSII-CARTAN-RATIONAL-SUBDIVISION-SMART-PATH-BOUNDARY

Ledger ID: R-105

Status: T4 ANALYTIC/EXACT/EXECUTED ADVANCE AND METHOD BOUNDARY.

Precise statement:

Complete common-root Cartan edges descend exactly to an extended-endpoint quotient. $F_{6.5}$ and K_R change sign under exact production subdivision while complete Delta W_R is invariant. Generic A9 monotonicity and the all-law pointwise relative bracket are false.

Scope: Fixed cutoff/floor; one Cartan root with complete heat edges; common rational target heat. Method no-gos are not Nelson counterexamples.

Dependencies:

R-085, R-088, R-092, R-093, R-098--R-104, A7, and A9.

Evidence grade:
 ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.

Reproduction command:
 E:\Dev\TECT.venv\Scripts\python.exe
 codes/foundations/
 a13_classii_cartan_rational_subdivision_smart_path_boundary_verify.py

Expected output:
 111/111 + 111/111 + 416/416 = 638/638 PASS; exit zero.

Negative results:
 RATIONAL-TAYLOR-OWNER-SUBDIVISION; GENERIC-SMART-PATH-MONOTONICITY;
 ALL-LAW-POINTWISE-RELATIVE-BRACKET; FULL-BUDGET-CRITICAL-YOUNG;
 ONE-PAIR-PRODUCT-FACTORIZATION
 (all prefixed NG-2026-07-28-A13- in the registry).

Falsification gate:
 Failure of the complete-heat Cartan quotient, normalized rational
 values or defect cancellation, top-shell harmonic/range-resolvent
 asymptotics, full-bracket order audit or $b(t) \geq 3/t$ consequence,
 one-pair/cross-mode identities, authority/hash/PDF/exploration/
 registry/count contracts, or any scope firewall.

Tier before / after:
 T4 / T4; no promotion.

No-overclaim statement:
 No distinct-root/visitwise theorem, Gibbs-specific/time-integrated
 bracket, OVERLAP_src, Nelson, removal, measure, T5--T7, or Sector-A
 closure is proved.

Next required action:
 Prove (11.1) by a complete signed packet or Gibbs/time-integrated A9.